AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2017

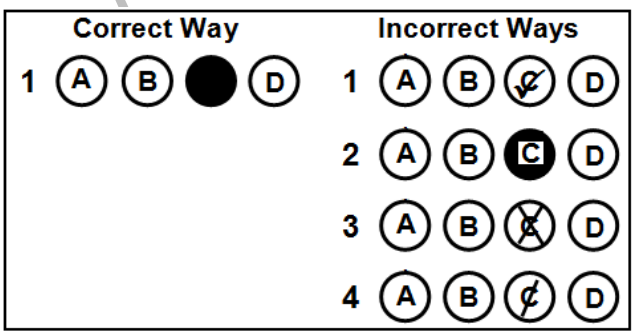
Biology Paper I

Time: 35 minutes Marks: 25

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INSTRUCTIONS

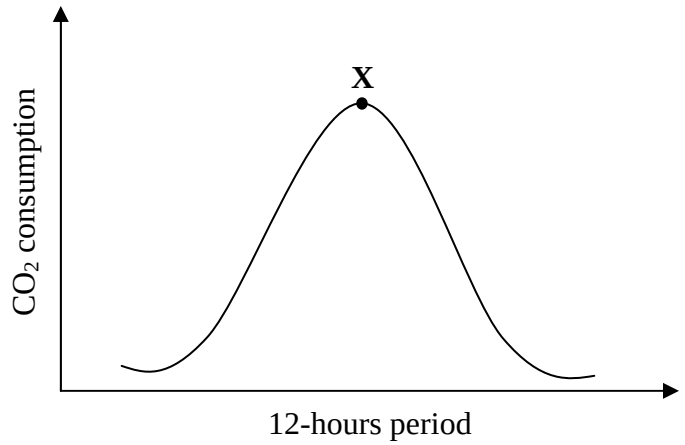
1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. The given graph shows the amount of CO_2 gas consumed by some plants in a garden at a maximum temperature of 35°C on a summer day.

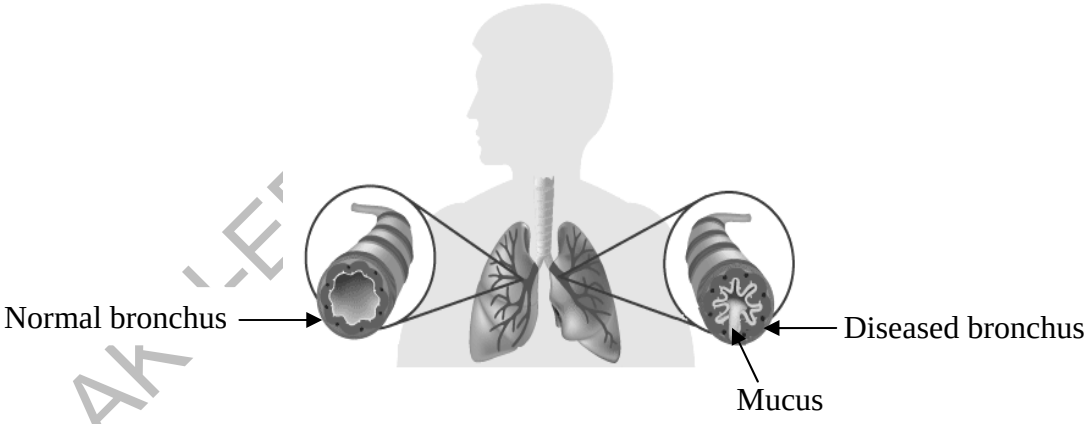


The process(es) occurring at point **X** is/ are

I. Gaseous Exchange
II. Photosynthesis
III. Cellular Respiration

A. I only
B. II only
C. II and III
D. I, II and III

2.



The diseased bronchus in the given diagram depicts symptoms of

A. cancer.
B. asthma.
C. pneumonia.
D. emphysema.

3. The excretion of concentrated urine in a marine fish is due to the presence of

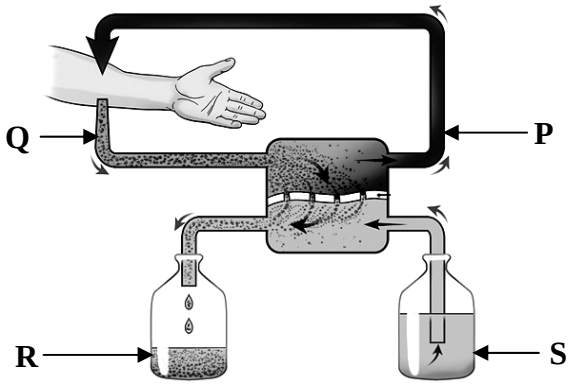
A. long loop of Henle.
B. distal convoluted tubule.
C. cup-shaped renal capsule.
D. proximal convoluted tubule.

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4. Which of the following steps are involved in the formation of urine in human beings?

	Selective Reabsorption	Excretion	Filtration
A	Yes	No	Yes
B	No	Yes	No
C	Yes	Yes	Yes
D	No	No	Yes

5. The process of haemodialysis is shown in the given diagram.



The concentration of urea would be the highest in

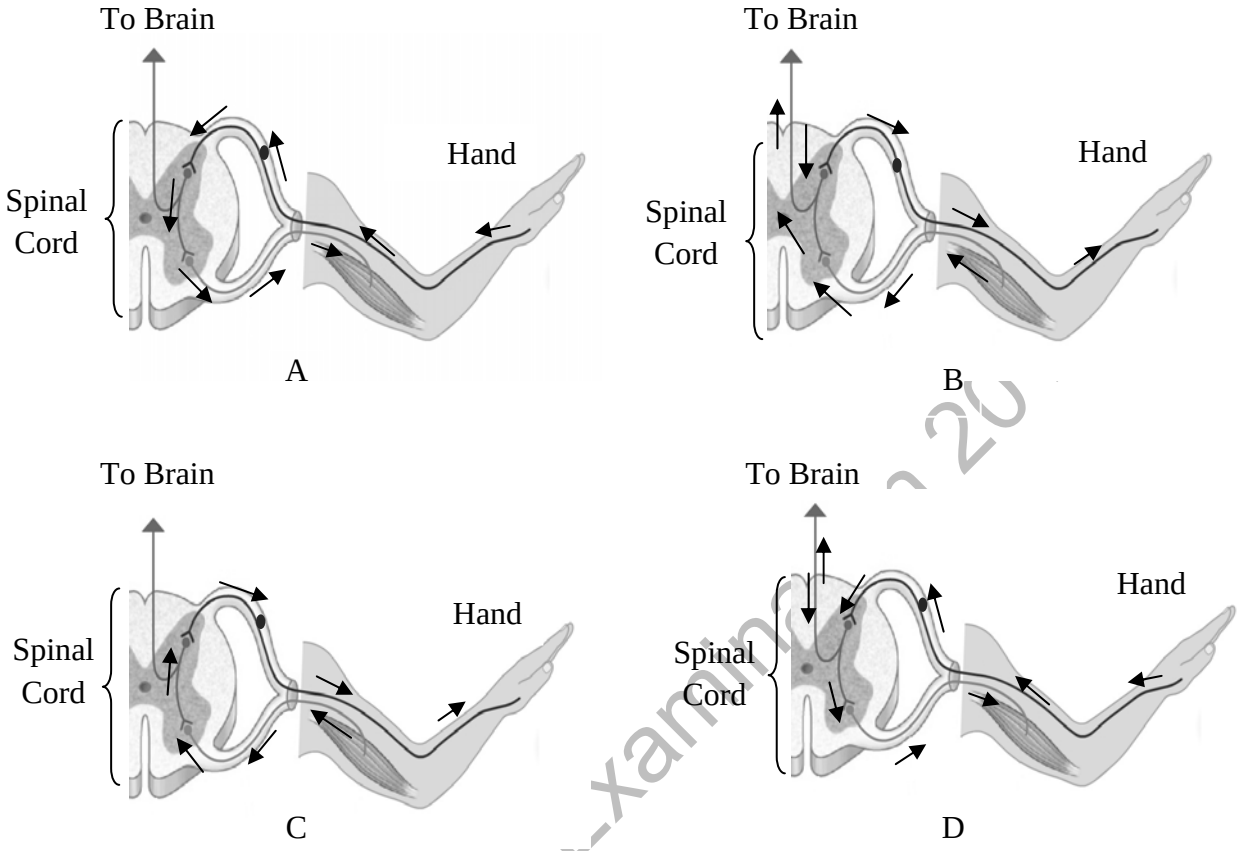
A. P and S.
B. Q and S.
C. P and R.
D. Q and R.

6. A patient is diagnosed with stones in his kidney. Which of the following methods of treatment should a doctor recommend to remove kidney stones?

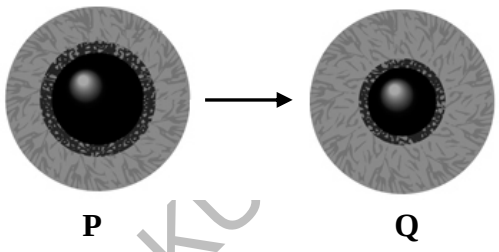
A. Lithotripsy
B. Haemodialysis
C. Blood transfusion
D. Peritoneal dialysis

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7. What will be the pathway for a spinal reflex action when the hand of an individual is pricked by a needle?



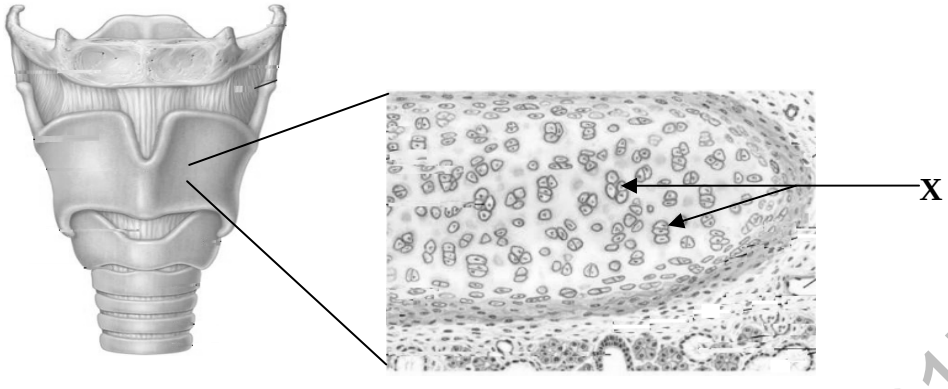
8. The given diagram shows change in the state of pupil of a person from **P** to **Q**.



The state **Q** shows that circular muscles of the iris have

A. relaxed as the person has entered a poorly lit room.
B. relaxed as the person has entered a brightly lit room.
C. contracted as the person has entered a poorly lit room.
D. contracted as the person has entered a brightly lit room.

9. The structures represented as **X** in the given diagram are



A. osteocytes.
B. lymphocytes.
C. chondrocytes.
D. thrombocytes.

10. Flexion of arm occurs when

	Biceps muscles	**Triceps muscles**
A	contract	relax
B	relax	contract
C	contract	contract
D	relax	relax

11. Osteoporosis characteristically occurs in elderly women due to the deficiency of

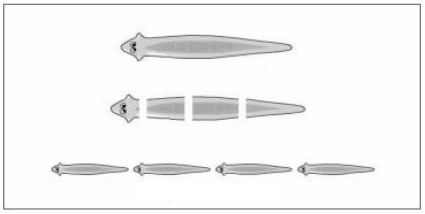
A. thyroxin.
B. oxytocin.
C. oestrogen.
D. calcitonin.

12. Double fertilisation in plants results in the formation of

A. one haploid zygote.
B. two haploid zygotes.
C. one triploid endosperm cell.
D. two triploid endosperm cells.

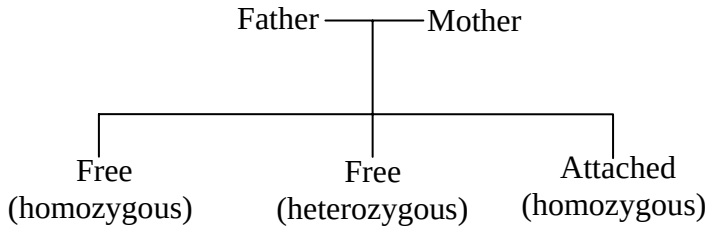
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13. The method of asexual reproduction shown in the given diagram is called



A. budding.
B. binary fission.
C. fragmentation.
D. spore formation.

14. In humans, free ear lobe is completely dominant over attached ear lobe. The given chart shows the pattern of inheritance of ear lobe in a family.



The genotypes of parents would be

	Mother	**Father**
A	Homozygous free	Heterozygous free
B	Heterozygous free	Heterozygous free
C	Homozygous free	Homozygous attached
D	Heterozygous free	Homozygous attached

15. Red flower colour is completely dominant over white flower colour in a particular plant.

When a heterozygous red flower is crossed with a white flower, the probable percentage of offspring flower colour would be

A. 100% red flowers.
B. 75% red and 25% white flowers.
C. 50% red and 50% white flowers.
D. 25% red and 75% white flowers.

16. Which of the following is an example of artificial selection?

A. Grapes are bred to produce large berry size.
B. Ladybirds are used to control aphids on crops.
C. Bacteria develop strains that are resistant to antibiotics.
D. Human gene is inserted into bacteria to produce insulin.

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17. The function of fungi in an ecosystem is to

    A. provide oxygen gas to producers.
    B. return nutrients to the environment.
    C. increase complexity of food chains.
    D. decrease competition among consumers.

18. The flow of material in an ecosystem is different from the flow of energy because it

    A. remains unaffected by sunlight.
    B. is cyclical through decomposers.
    C. is not returnable to the environment.
    D. moves from producers to consumers.

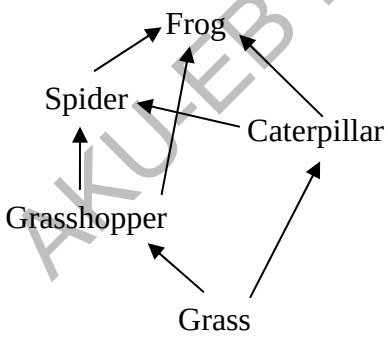
19. Following are the steps of the nitrogen cycle.

    I. Conversion of ammonia into nitrates
    II. Formation of proteins
    III. Conversion of N_2 gas to NH_3 gas
    IV. Conversion of nitrates into N_2 gas

    Which of the following is the CORRECT sequence of events of nitrogen cycle?

    A. III, I, II, IV
    B. I, II, IV, III
    C. I, IV, III, II
    D. III, II, IV, I

20. Which of the following animals fall in the category of both secondary and tertiary consumers in the given food web?



    A. Frog
    B. Spider
    C. Caterpillar
    D. Grasshopper

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21. Students of 10th grade in a particular school are following the '3R' principle of environmental care.

    Which of the following is an example of 'reduce'?

    A. Storing spices in empty bottles of jam.
    B. Composting kitchen waste in the garden.
    C. Avoiding giving presents with excess packaging.
    D. Repairing broken items rather than buying new ones.

22. In an industrial fermenter, an impeller is a device which

    A. controls pH of the culture broth.
    B. mixes air bubbles with nutrient broth.
    C. removes heat during microbial activities.
    D. monitors temperature inside the fermenter.

23. A bacterium is considered genetically modified when it has

    A. a foreign gene.
    B. many plasmids.
    C. a complete genome.
    D. restriction endonucleases.

24. Penicillin is a drug obtained from a/ an

    A. plant.
    B. animal.
    C. fungus.
    D. bacterium.

25. Which of the given properties describe vaccines?

    I. Contain dead pathogens
    II. Used to treat bacterial diseases
    III. Stimulate the immune response

    A. I only
    B. I and II
    C. I and III
    D. II and III

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END OF PAPER